

The Engineering Behind Modern Databases: B-Trees, Wal, Mvcc And Distributed Storage

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Storage

Half-title and publisher imprint.

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SYSTEM PRINCIPLE

Deterministic state transitions ensure reproducibility across replicas regardless of message arrival interleaving.

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CHAPTER 1

Foundations and Core Concepts

Content for section 'Historical Context & Evolution' focusing on timeline.

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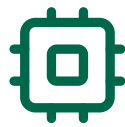
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CHAPTER 2

Architectural Principles

Content for section 'System Topology & Component Model' focusing on diagram.

The write path ensures zero data loss by recording mutations to disk before acknowledging client requests.

```
graph TD
    Client[Client Write] -->|1. Write| WAL[(Write-Ahead Log)]
    Client -->|2. Insert| MemTable[In-Memory MemTable]
    MemTable -->|3. Flush Threshold| Immutable[Immutable MemTable]
    Immutable -->|4. Background Flush| L0[Level 0 SSTable]
    L0 -->|5. Compaction| L1[Level 1 SSTables]
```

Figure 1.1: LSM-Tree Ingestion & Compaction Pipeline

Content for section 'Core Protocols & State Machines' focusing on text.

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CHAPTER 3

Practical Implementation

Content for section 'Reference Implementation Patterns' focusing on code.

The consensus engine coordinates state replication across clustered nodes via atomic term transitions.

consensus/raft.rs

RUST

```
pub struct RaftNode {
    pub node_id: String,
    pub current_term: u64,
    pub voted_for: Option<String>,
    pub log: Vec<LogEntry>,
    pub commit_index: usize,
}

impl RaftNode {
    pub async fn start_election(&mut self) -> Result<bool, ElectionError> {
        self.current_term += 1;
        self.voted_for = Some(self.node_id.clone());
        let votes = self.broadcast_request_votes().await?;
        Ok(votes > self.peers.len() / 2)
    }
}
```

Listing 1.1: Raft election logic and term advancement.

○ IMPLEMENTATION DETAIL

Always persist `voted_for` and `current_term` to non-volatile WAL storage prior to acknowledging RPC vote requests.

Content for section 'Integration & Developer Experience' focusing on table.

Trade-off matrix evaluating storage architectures under high-throughput workloads:

Table 1.1: Architectural trade-offs between B+ Trees and LSM Trees.

CHARACTERISTIC	B+ TREE INDEX	LSM-TREE STORAGE ENGINE
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CHAPTER 4

Performance, Benchmarks & Trade-offs

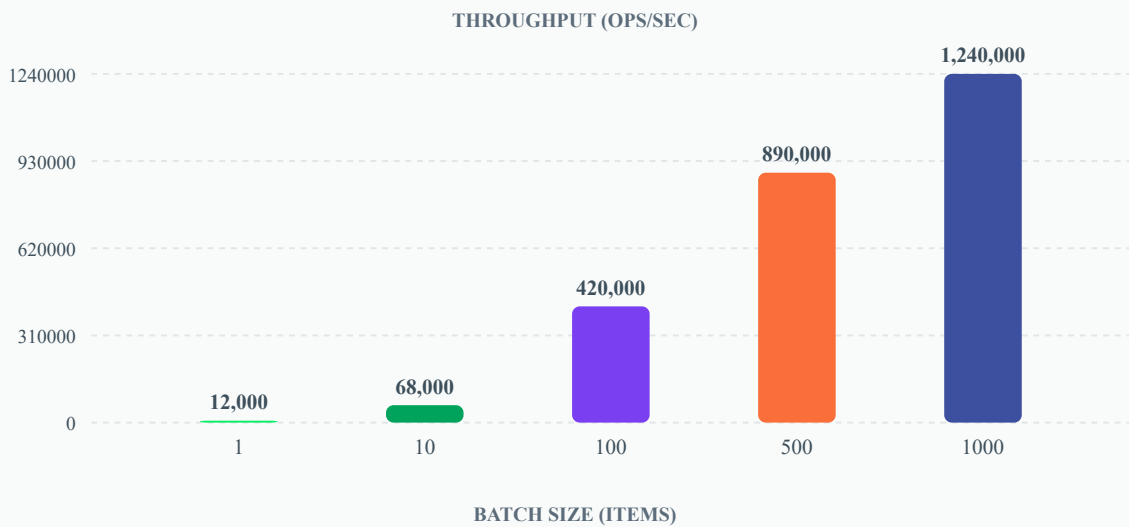
Content for section 'Throughput & Latency Benchmarks' focusing on statistic.

1.24M

OPERATIONS PER SECOND

Sustained write throughput benchmarked across a 64-node distributed NVMe cluster.

Write Throughput Scaling by Batch Size



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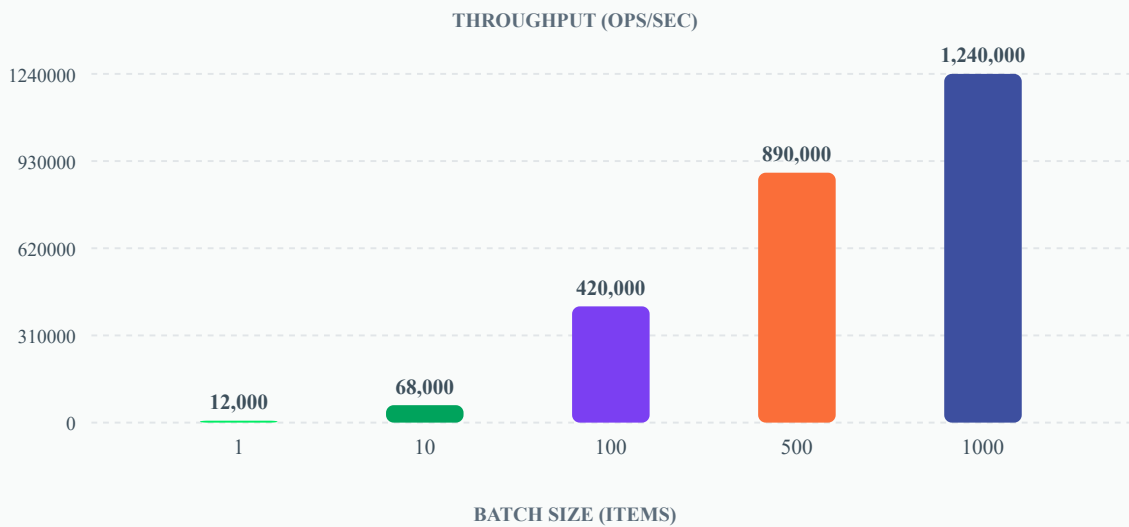
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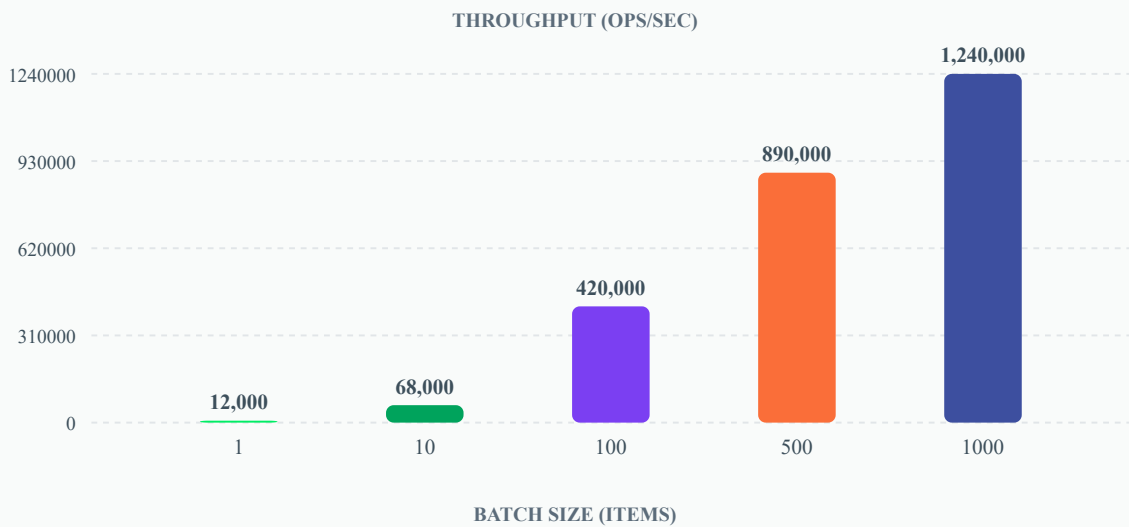
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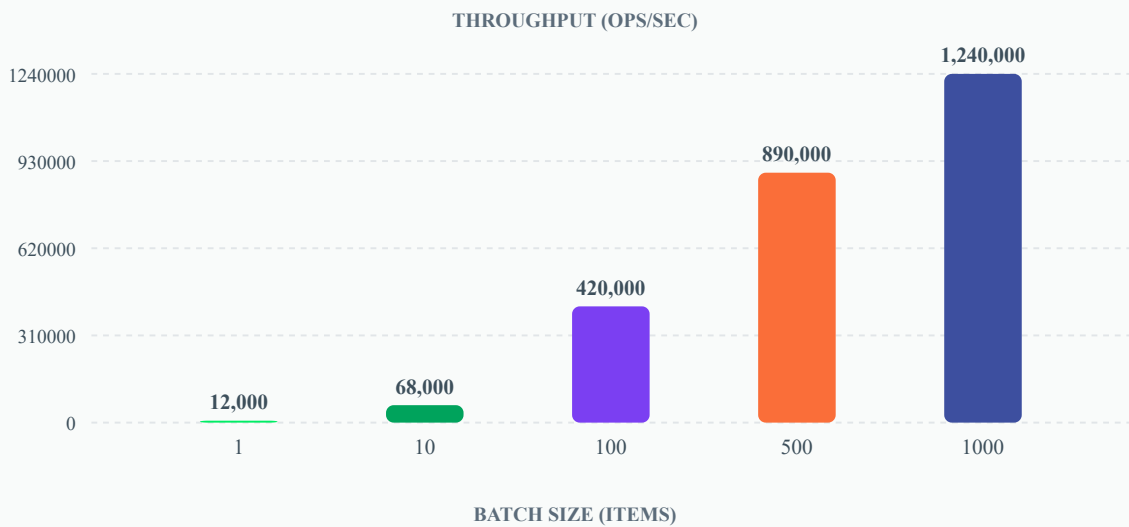
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CHAPTER 5

Production Best Practices & Case Studies

Content for section 'Scaling & Reliability Strategies' focusing on quote.

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